



CSE 4621: Microprocessors and Interfacing

Lecture 3 — 8085 Microprocessor Internal Architecture

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Lecture Plan

The agenda for today

- Intel 8085 Architecture Diagram
- Intel 8085 Pin Diagram
- Registers of Intel 8085
- Simple Assembly Program for Intel 8085

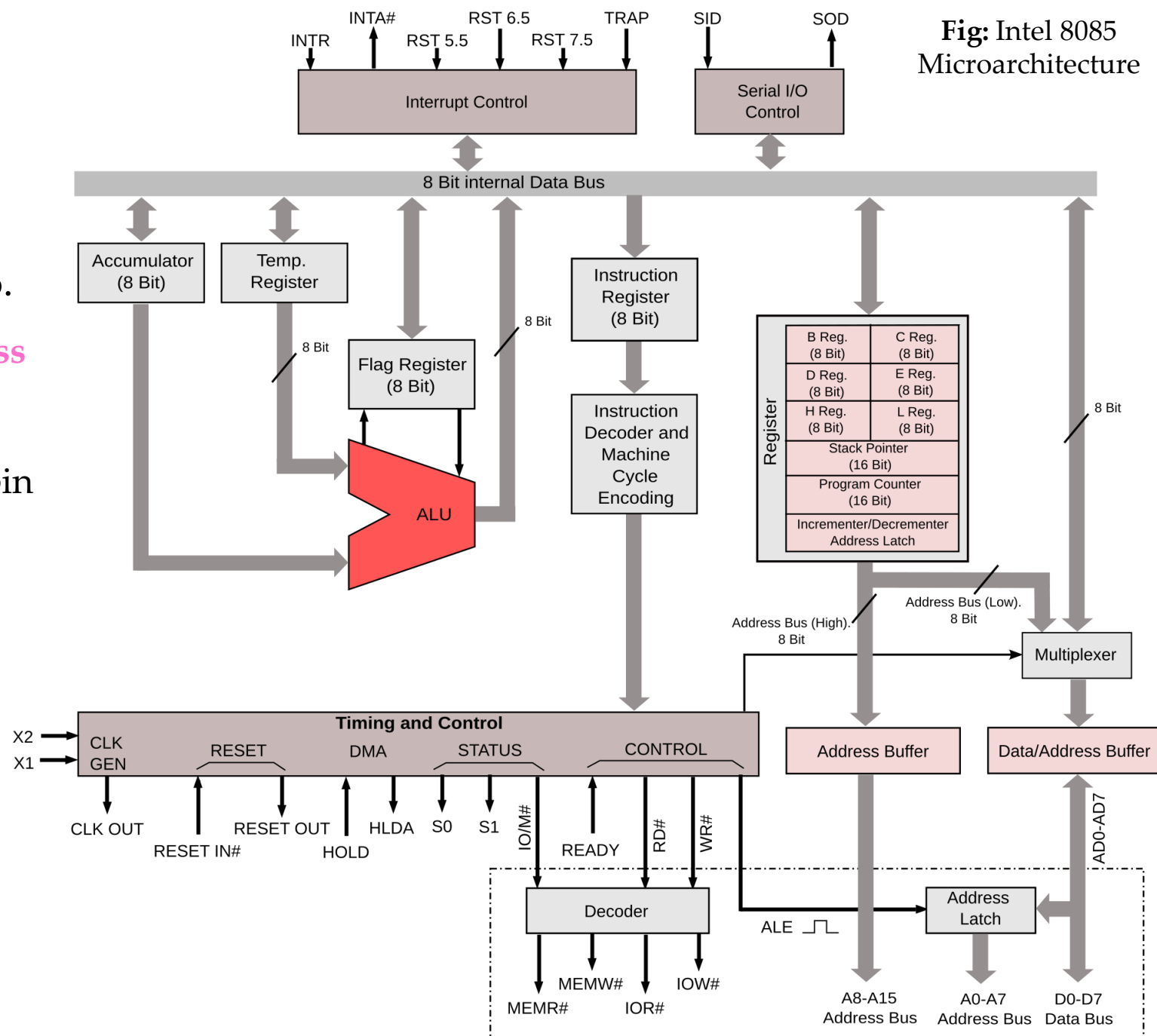
Intel 8085

What is it?

- The 8085 microprocessor was introduced by Intel in the year 1976.
- **8-bit** microprocessor with **16-bit address bus** and **8-bit data bus**.
- It is an 8-bit microprocessor with a 40 pin *Dual In-line Package (DIP)*.
- Total **74 operation codes** in assembly language and those can generate **246 instructions**.

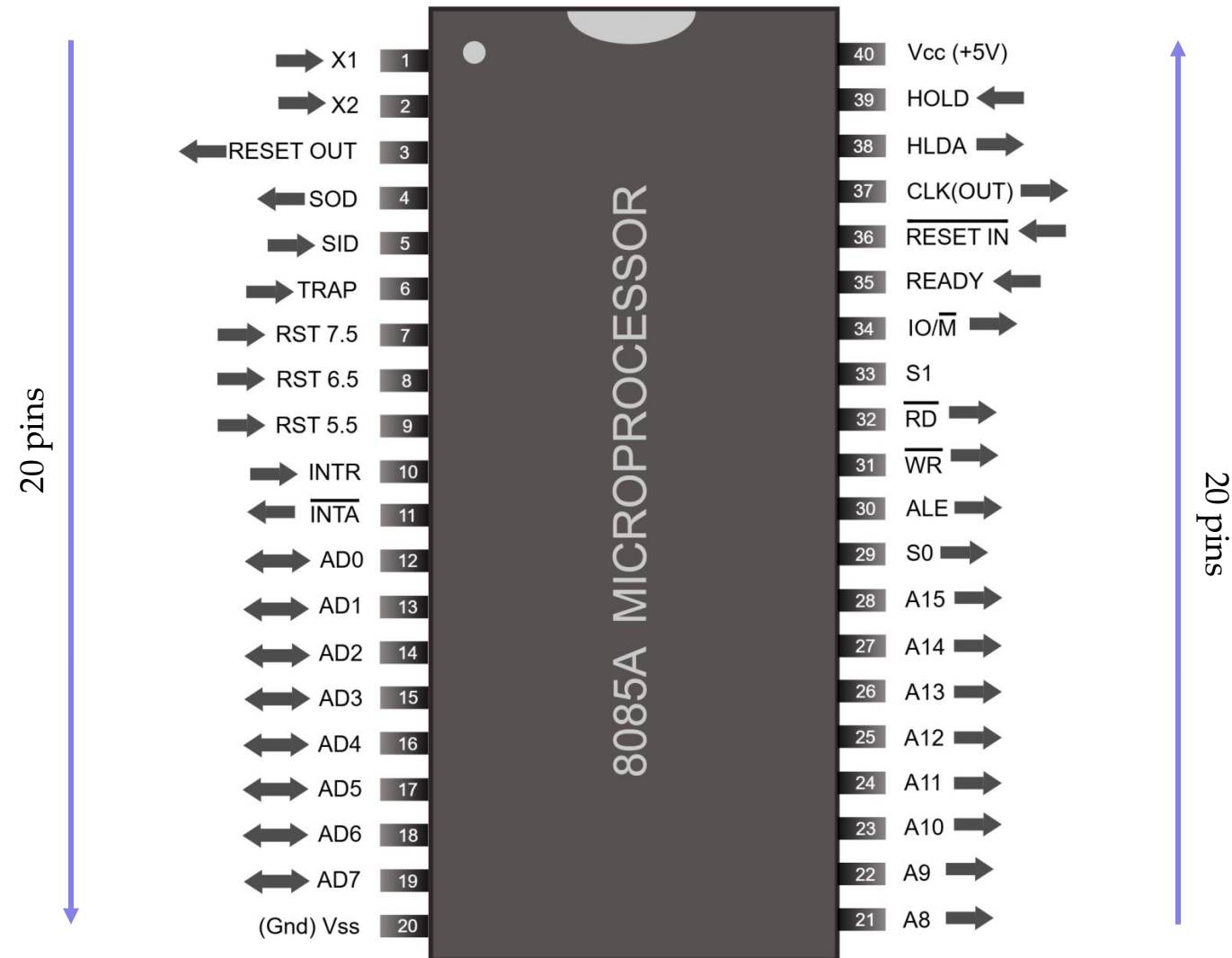


Fig: Intel P8085AH-2



Intel 8085 Pin Diagram

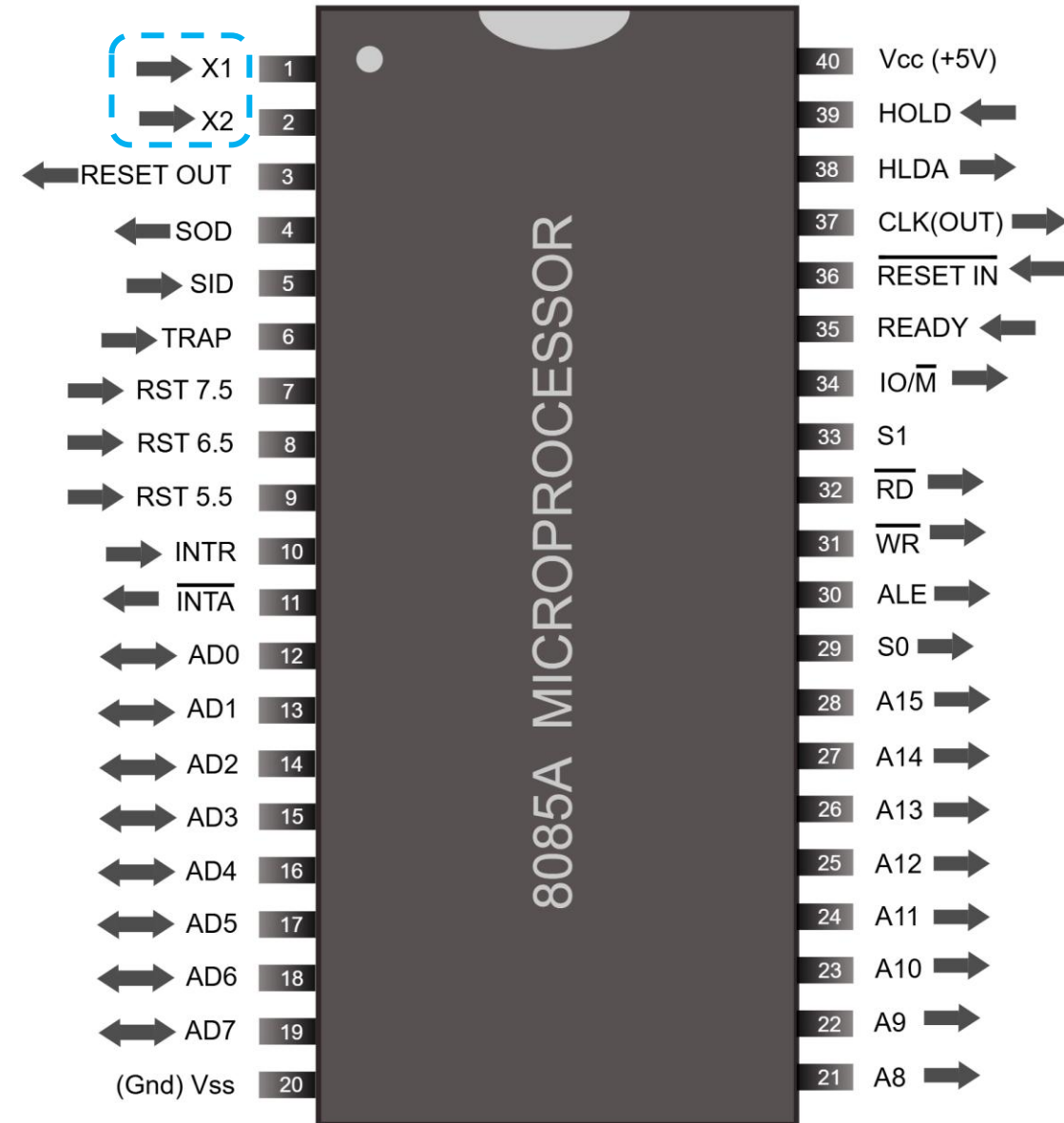
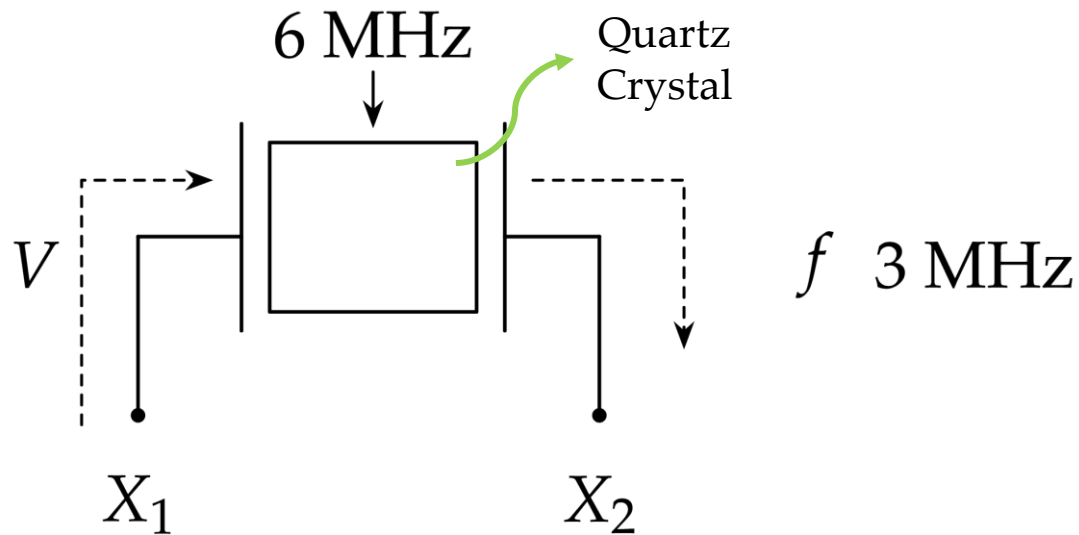
What does it look like?



Intel 8085 Pins

X_1 and X_2

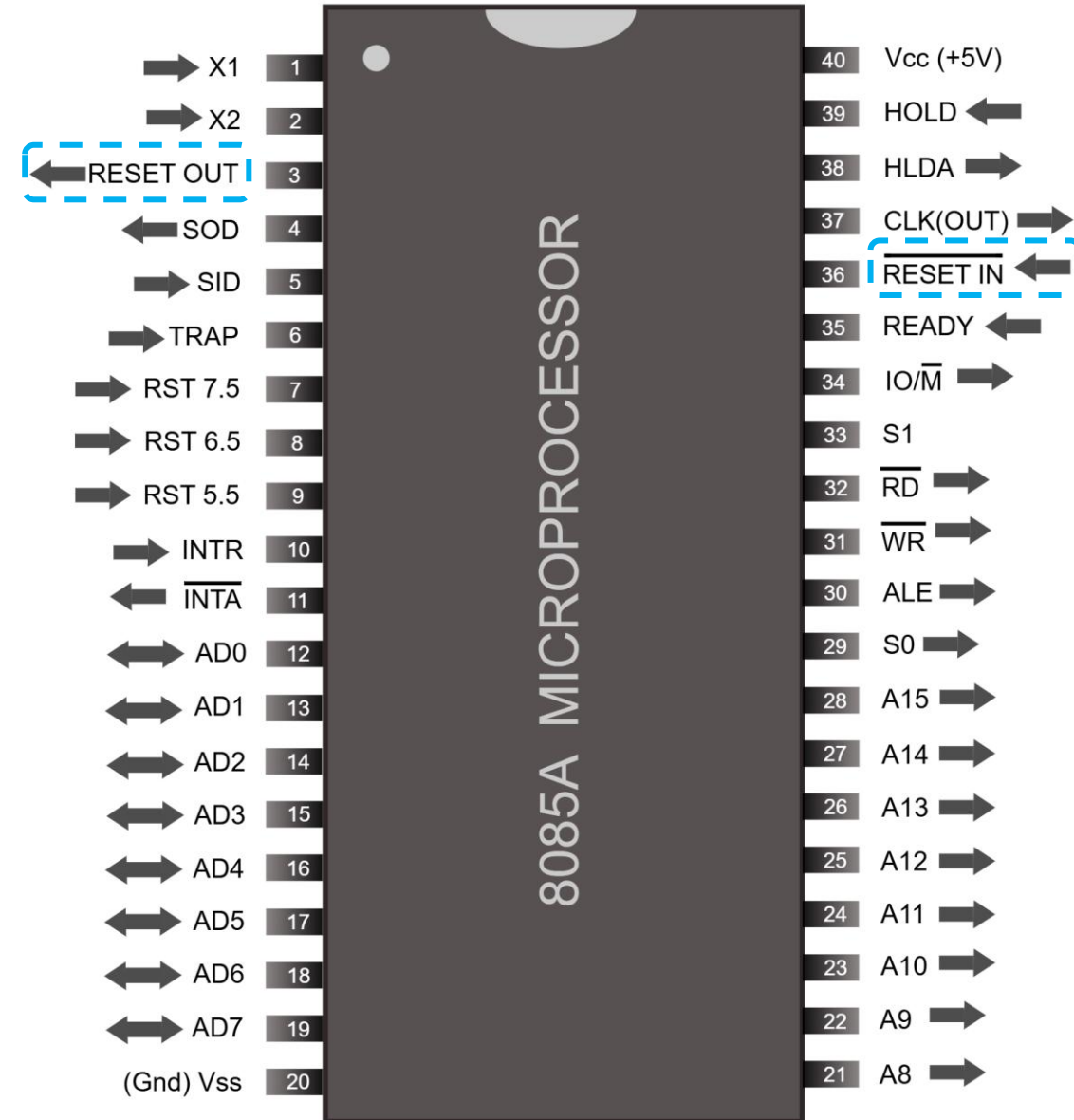
- Piezoelectric Crystal or R/C network connections to set the internal clock generator.
- X_1 can also be an external clock input instead of a crystal.
- The input frequency is divided by 2 to give the internal operating frequency.



Intel 8085 Pins

RESET OUT and $\overline{\text{RESET IN}}$

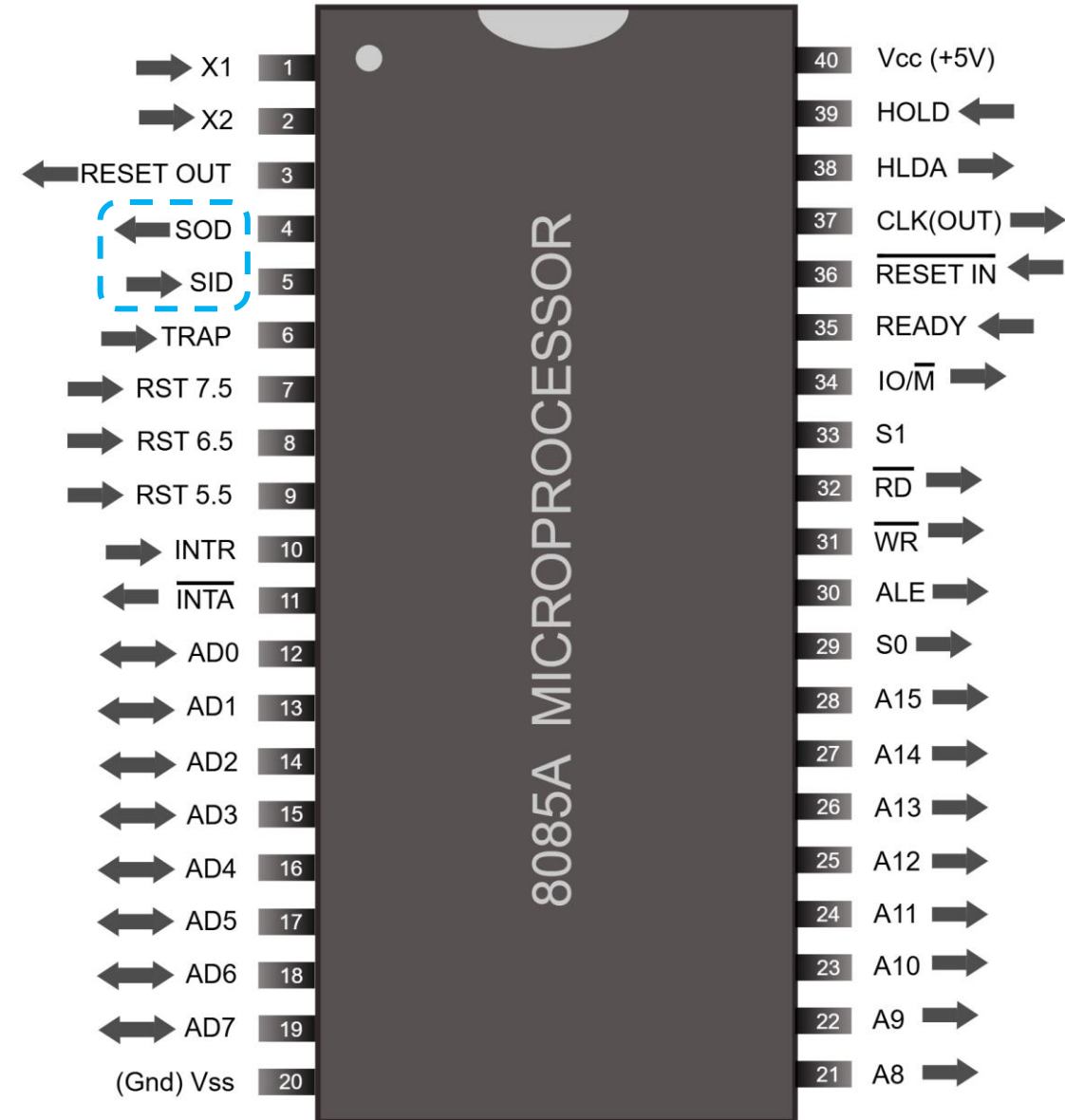
- RESET OUT is used to reset the **peripherals** connected to the microprocessor.
- $\overline{\text{RESET IN}}$ sets the **Program Counter** to 0 and clears the **buses**.
(signal needs to persist for *~0.5 microseconds*)



Intel 8085 Pins

SOD and SID

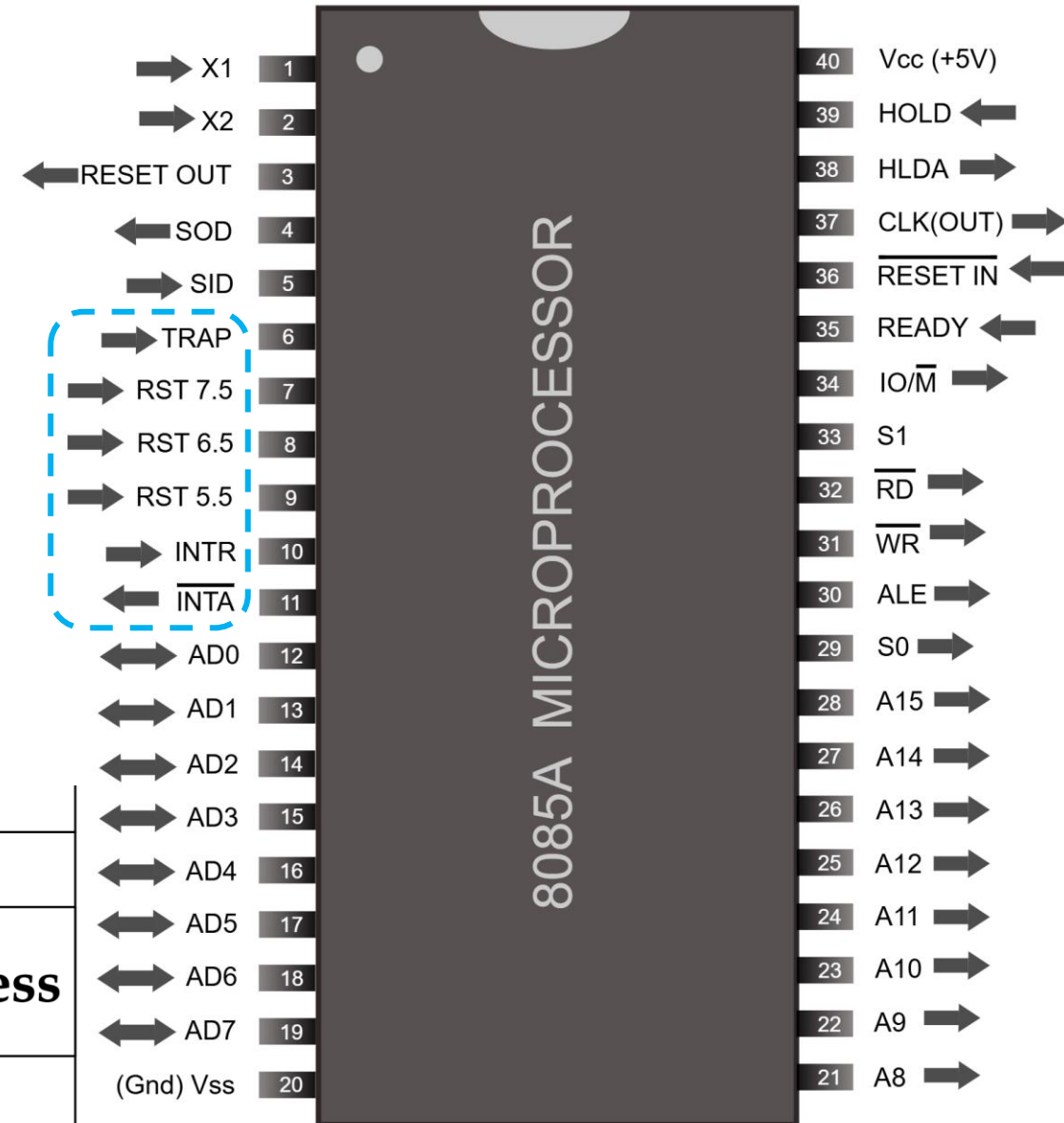
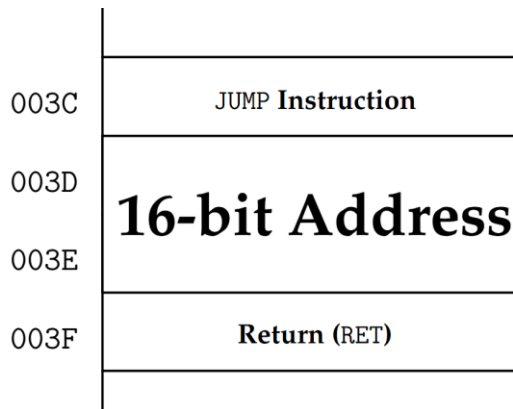
- Serial Output Data (SOD)
 - ✓ Outflux of serialized data **from** the μP
- Serial Input Data (SID)
 - ✓ Influx of serialized data **to** the μP



Intel 8085 Pins

Interrupt Pins

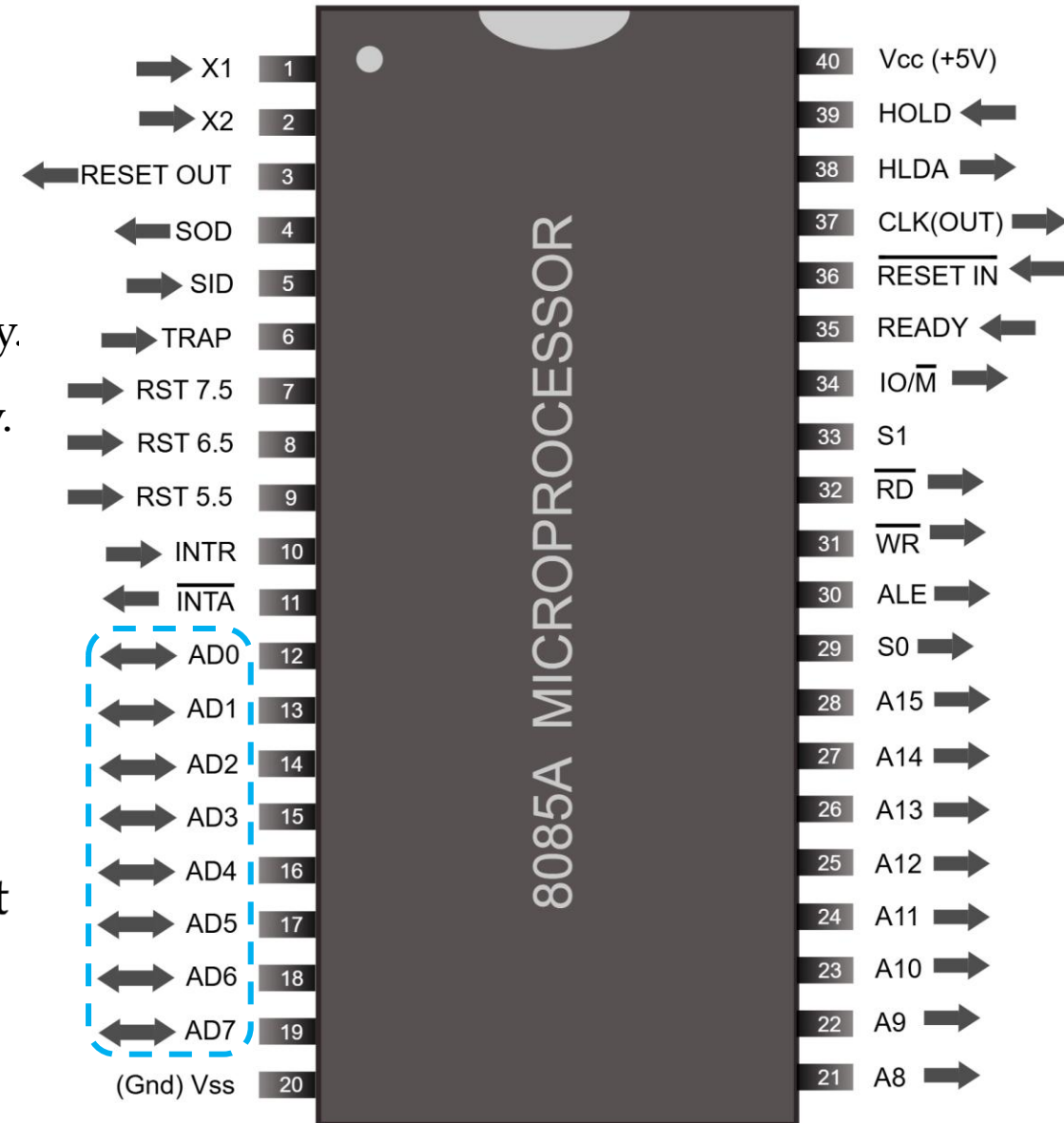
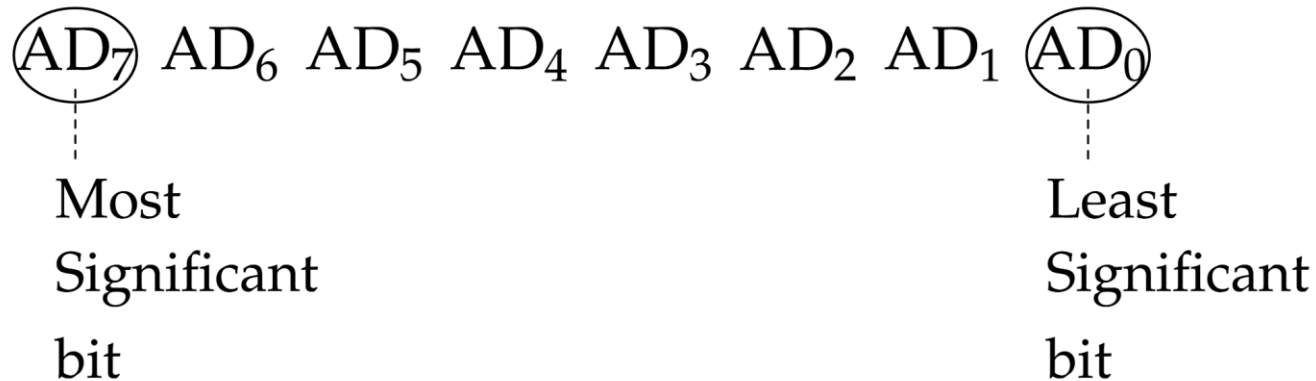
- Trap Interrupt (TRAP)
 - ✓ Non-maskable interrupt
 - ✓ Generated by external device
 - ✓ Its vector address is **0024h**
- Restart Interrupts (RST 7.5/6.5/5.5)
 - ✓ Maskable interrupt
 - ✓ Generated by software to handle exceptions/errors
 - ✓ Priority: RST 7.5 > RST 6.5 > RST 5.5
 - ✓ Vector addresses:
 - $7.5 \times 8 = 60$ which is **003Ch**
 - $6.5 \times 8 = 52$ which is **0034h**
 - $5.5 \times 8 = 44$ which is **002Ch**
- Interrupt Request/Acknowledgement (INTR/INTA)
 - ✓ Maskable interrupt
 - ✓ Generated by external devices



Intel 8085 Pins

Data Bus

- Intel 8085's data bus is *8 bits* long.
- Used for receiving the program code from memory.
- Used for receiving a data byte from an input port or memory.
- Used for sending out data byte to an output port or memory.

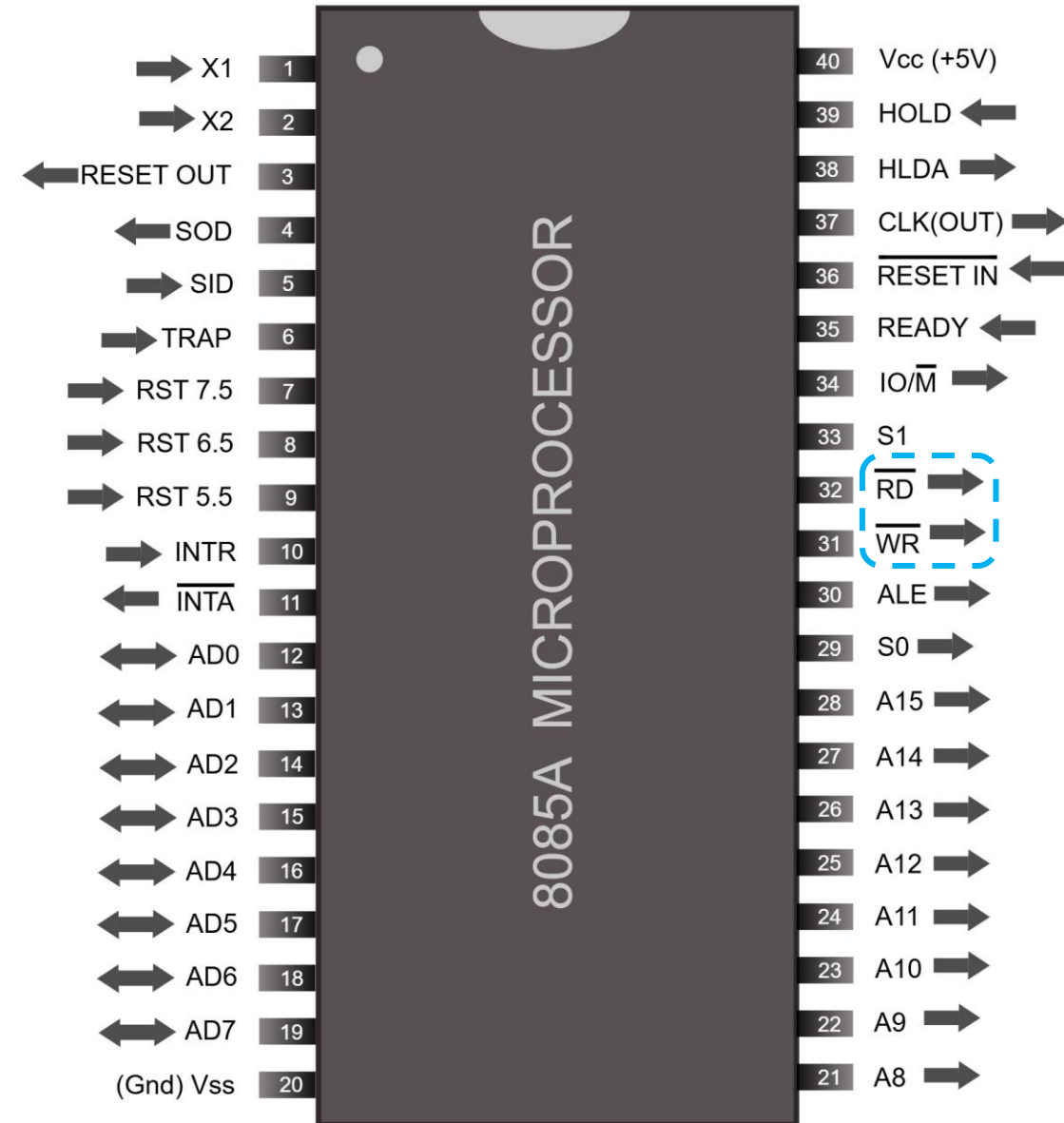


Intel 8085 Pins

$\overline{WR}$ and $\overline{RD}$

- Used for writing and reading, respectively.

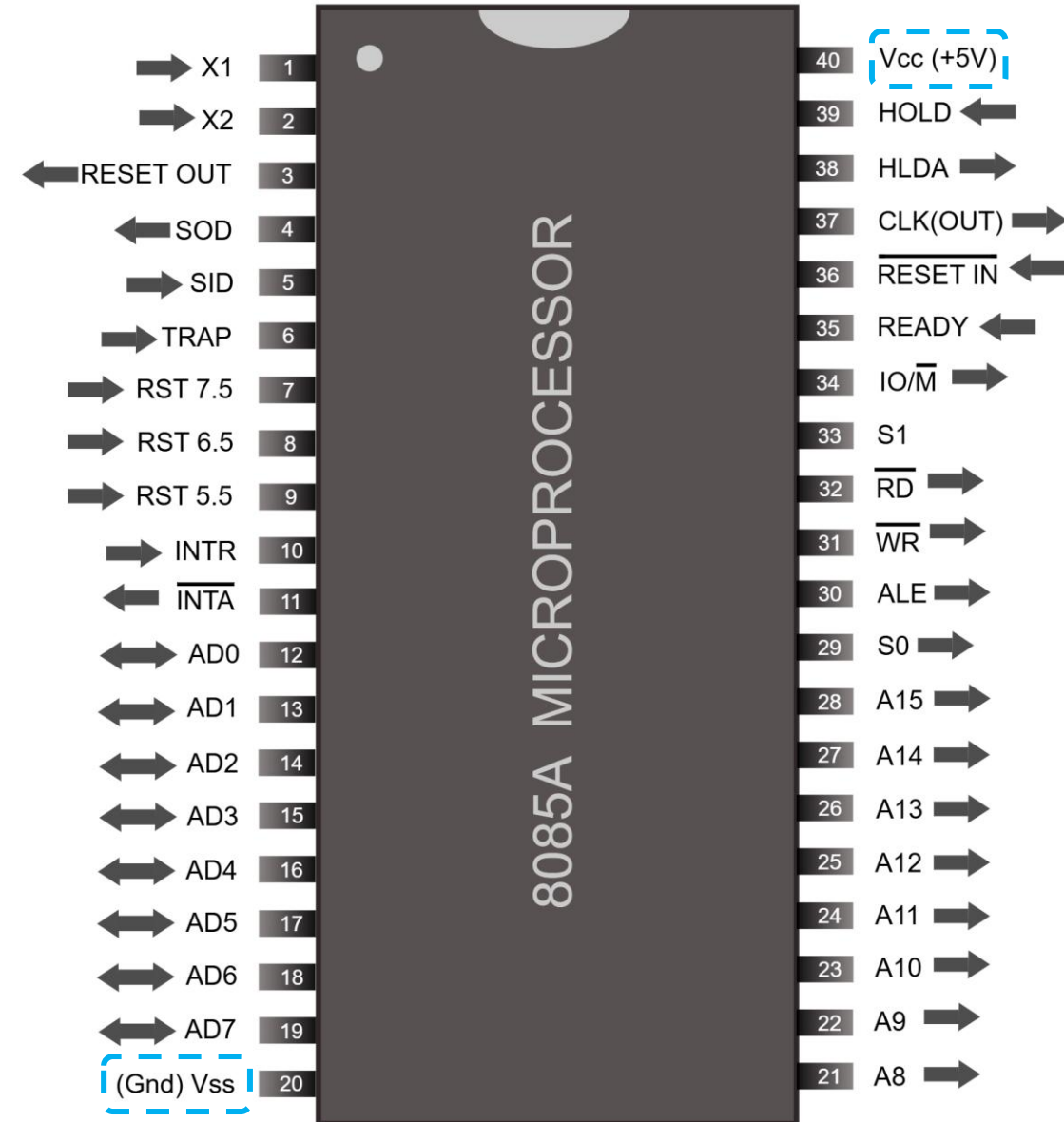
$\overline{WR}$	$\overline{RD}$	Function
0	1	AD_{7-0} are output pins.
1	0	AD_{7-0} are input pins.
1	1	Internal processing.
0	0	μP has malfunctioned.



Intel 8085 Pins

V_{CC} and V_{SS}

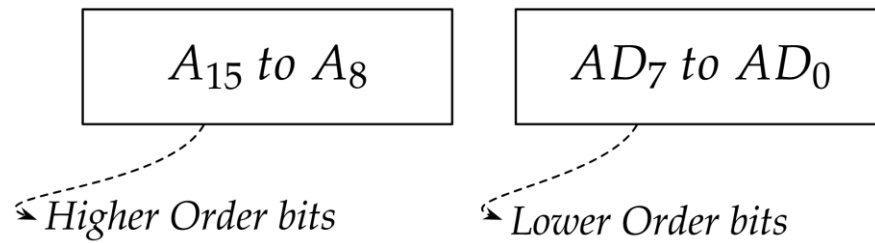
- $V_{CC}(+5V)$ is used to provide a +5V DC power supply.
- $V_{SS}(GND)$ grounds the IC.



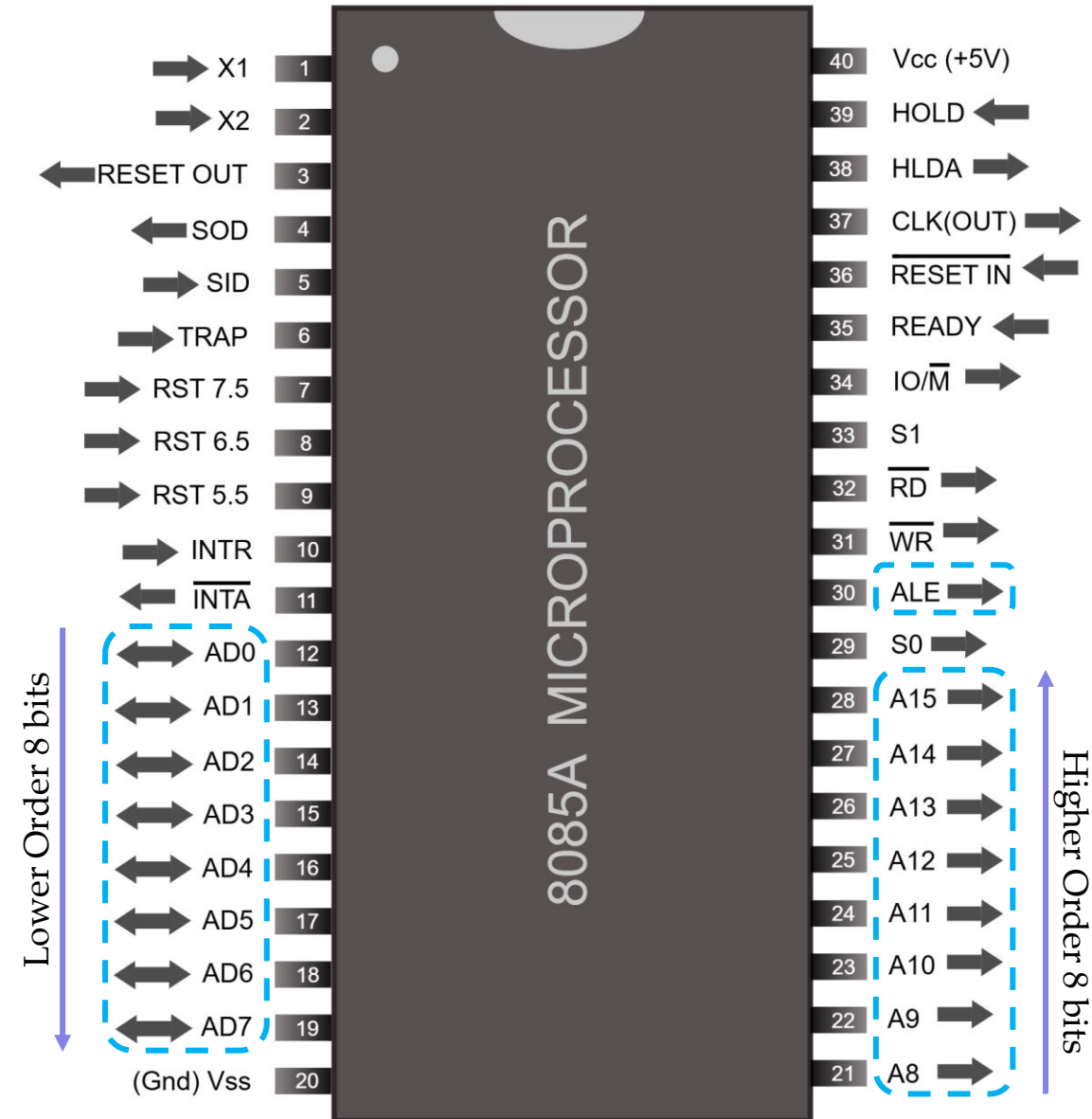
Intel 8085 Pins

Address Bus and ALE

- Intel 8085's address bus is **16 bits** long.
- Can be used to reach 65,536 different memory locations.



- Address Latch Enable (ALE)
 - ✓ When **1**, all **16 bits** are used by Address Bus.
 - ✓ When **0**, the **lower 8 bits** are used by the Data Bus.

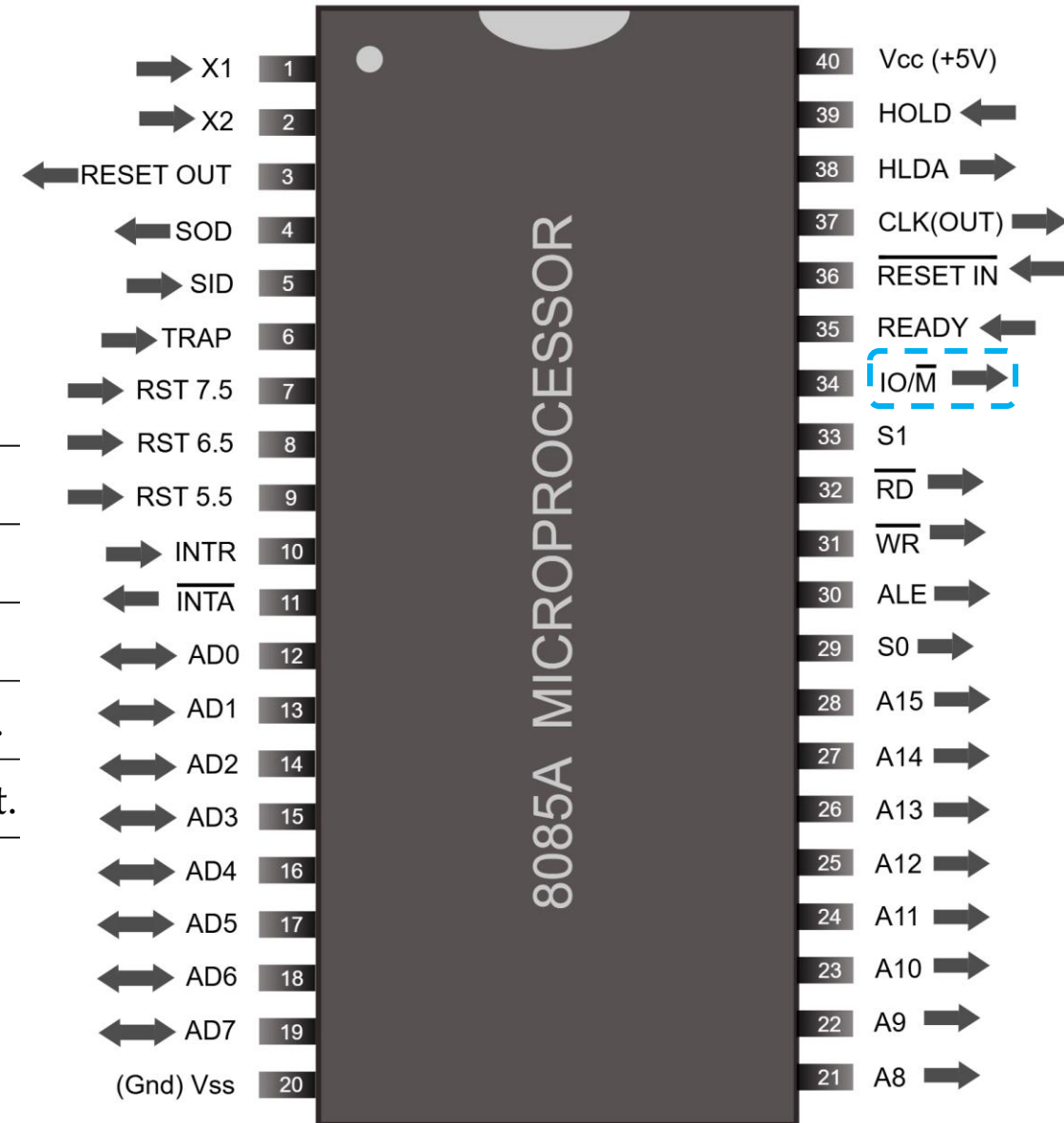


Intel 8085 Pins

IO/ $\overline{M}$ pin for I/O Devices and Memory

- It indicates whether the Read/Write is to memory or I/O device.

IO/ $\overline{M}$	$\overline{WR}$	$\overline{RD}$	Function
0	0	1	μP writes to memory.
0	1	0	μP reads from memory.
1	0	1	μP writes to output port.
1	1	0	μP reads from input port.

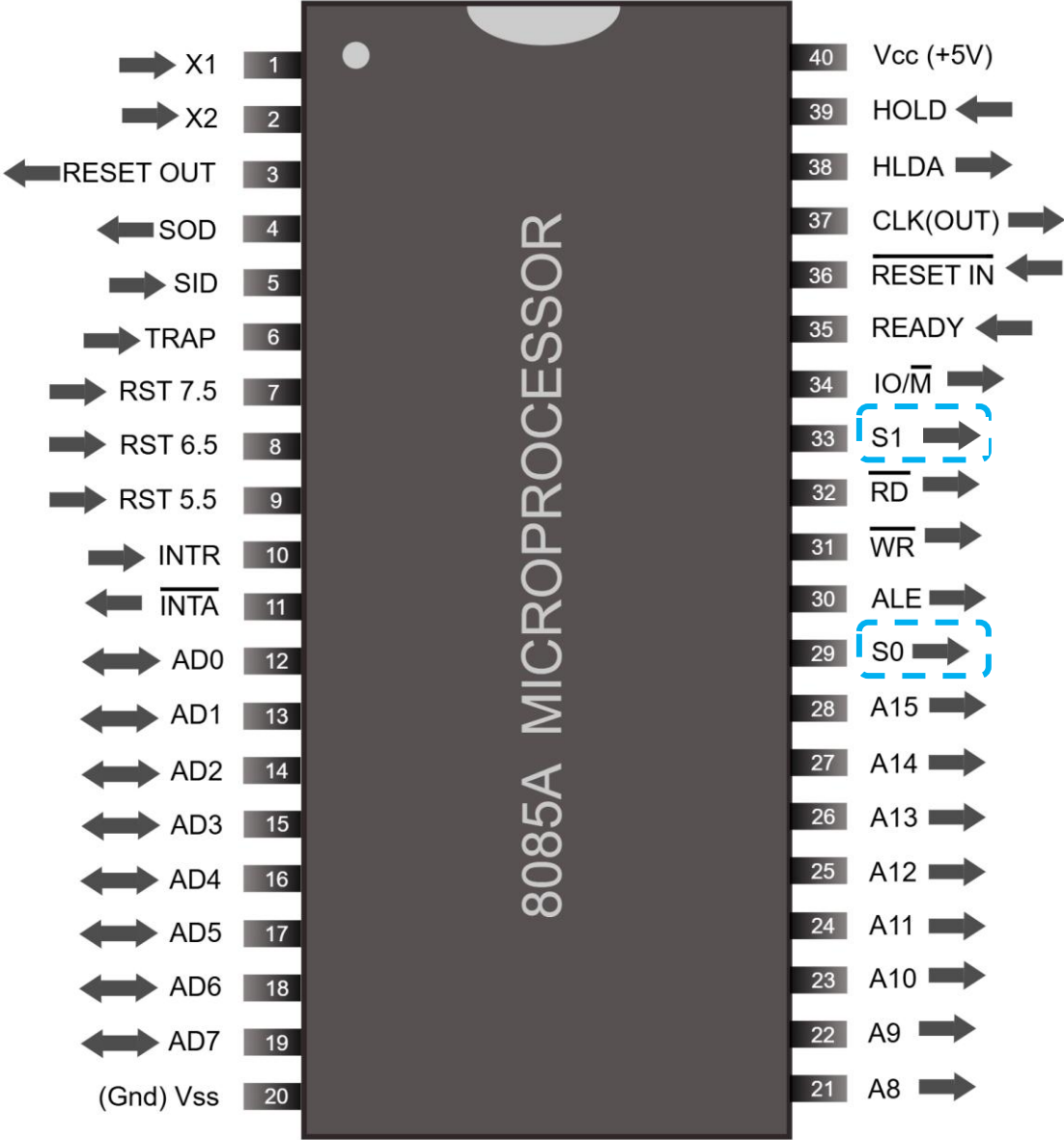


Intel 8085 Pins

S₀ and S₁

- These are pins for Status signals.
- Indicate different phases of the Instruction Cycle.

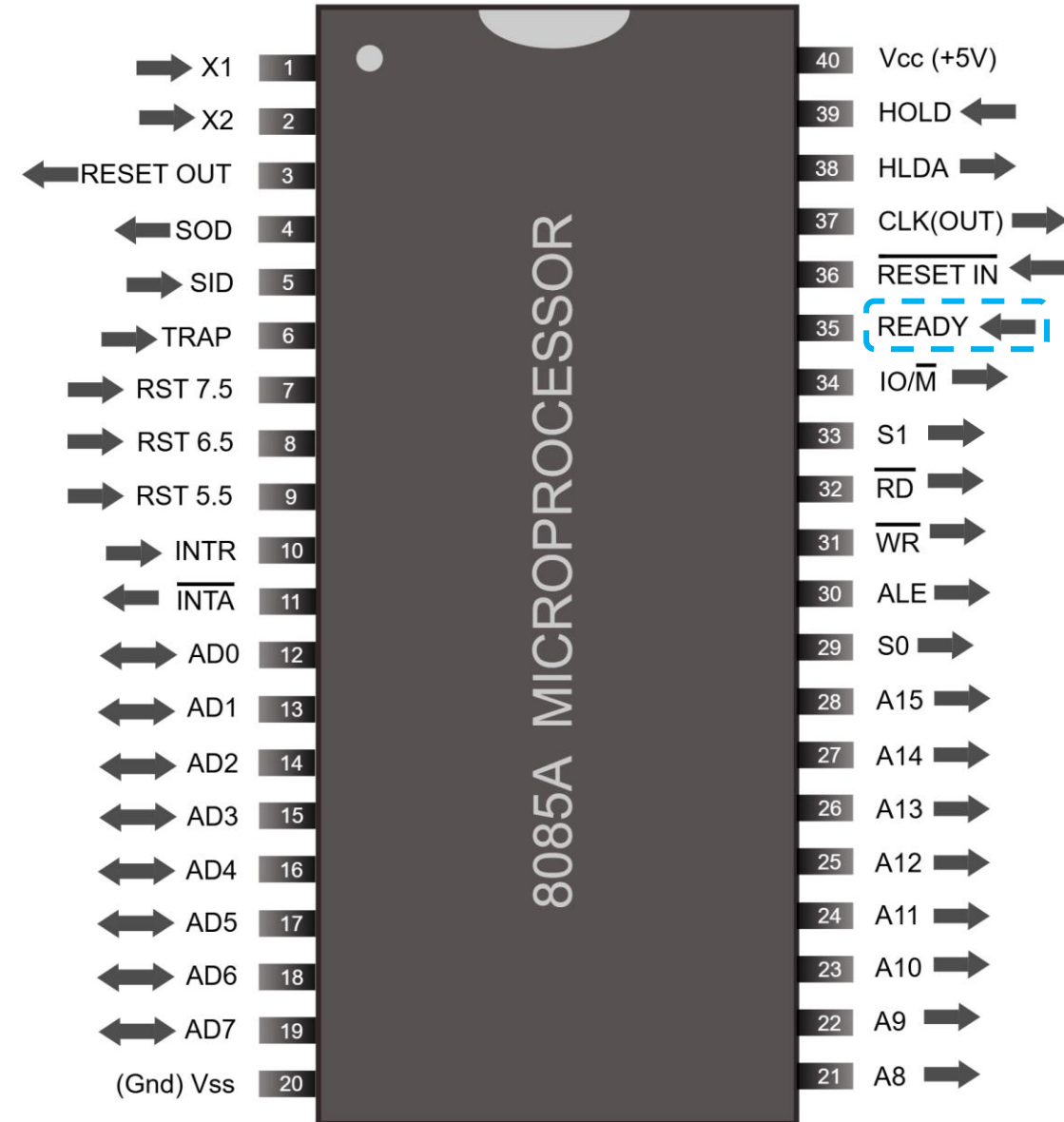
IO/ $\overline{M}$	S ₁	S ₀	Function
0	1	1	Opcode Fetch (OF)
0	1	0	Memory Read (MR)
0	0	1	Memory Write (MW)
1	1	0	I/O Read (IOR)
1	0	1	I/O Write (IOW)
1	1	1	Interrupt Acknowledge (INA)



Intel 8085 Pins

READY pin

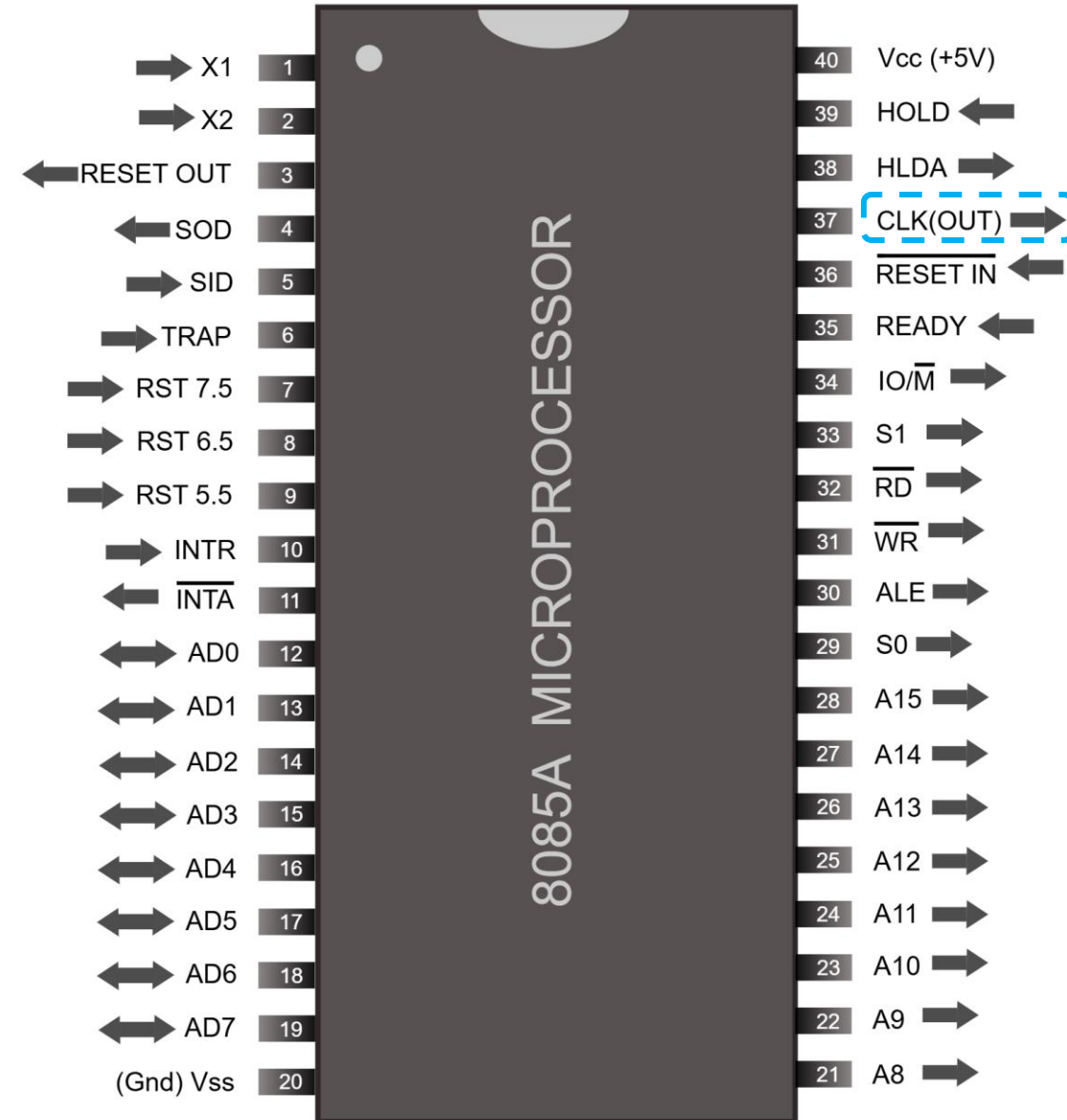
- Indicates whether an I/O device is ready to send/receive data.
 - ✓ When **1**, the I/O device is ready.
 - ✓ When **0**, the μP is waiting.



Intel 8085 Pins

CLK (OUT) pin

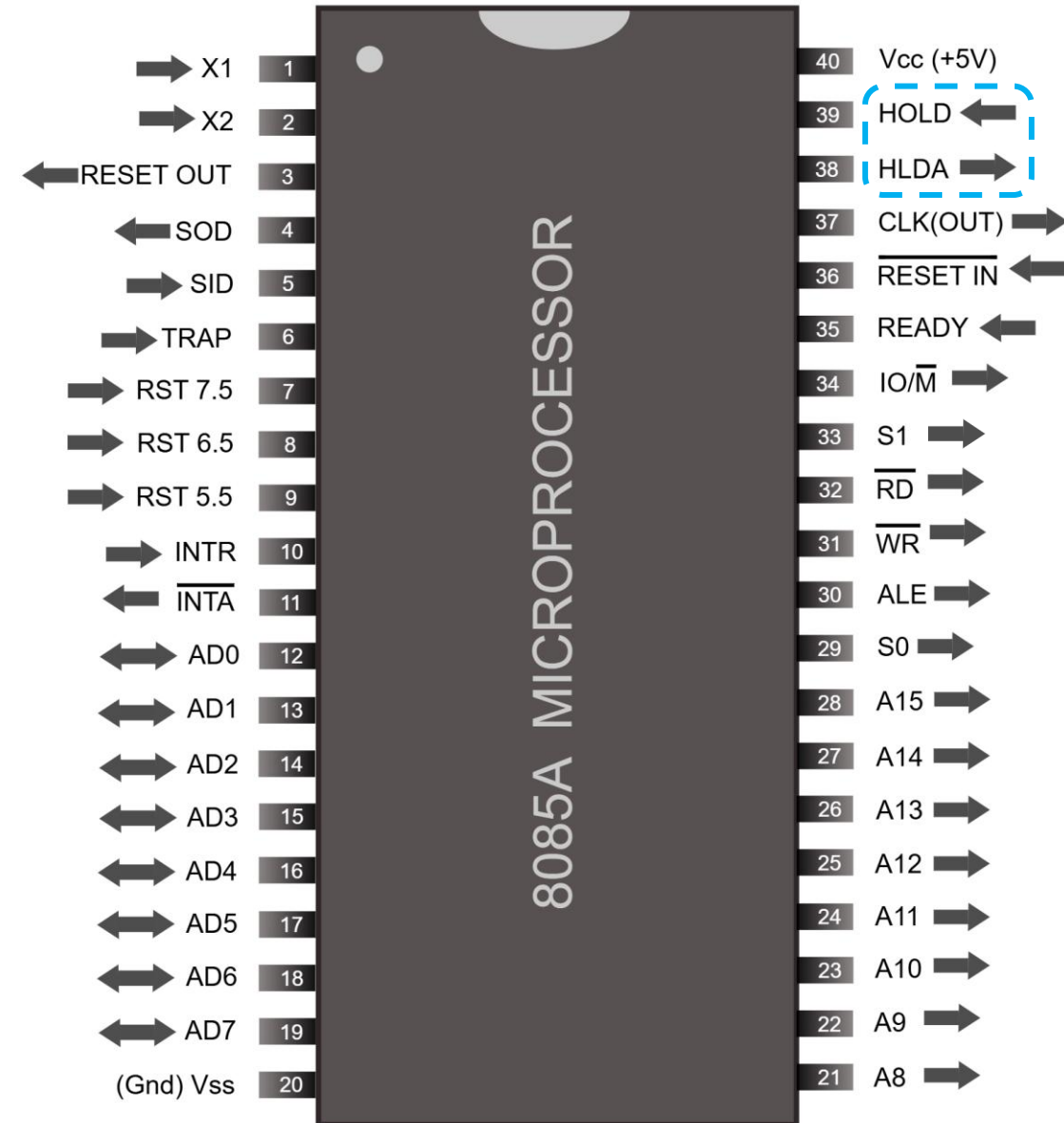
- Provides a timing reference for various components.
 - ✓ Helps attached peripherals synchronize with the μP .



Intel 8085 Pins

HOLD and HLDA

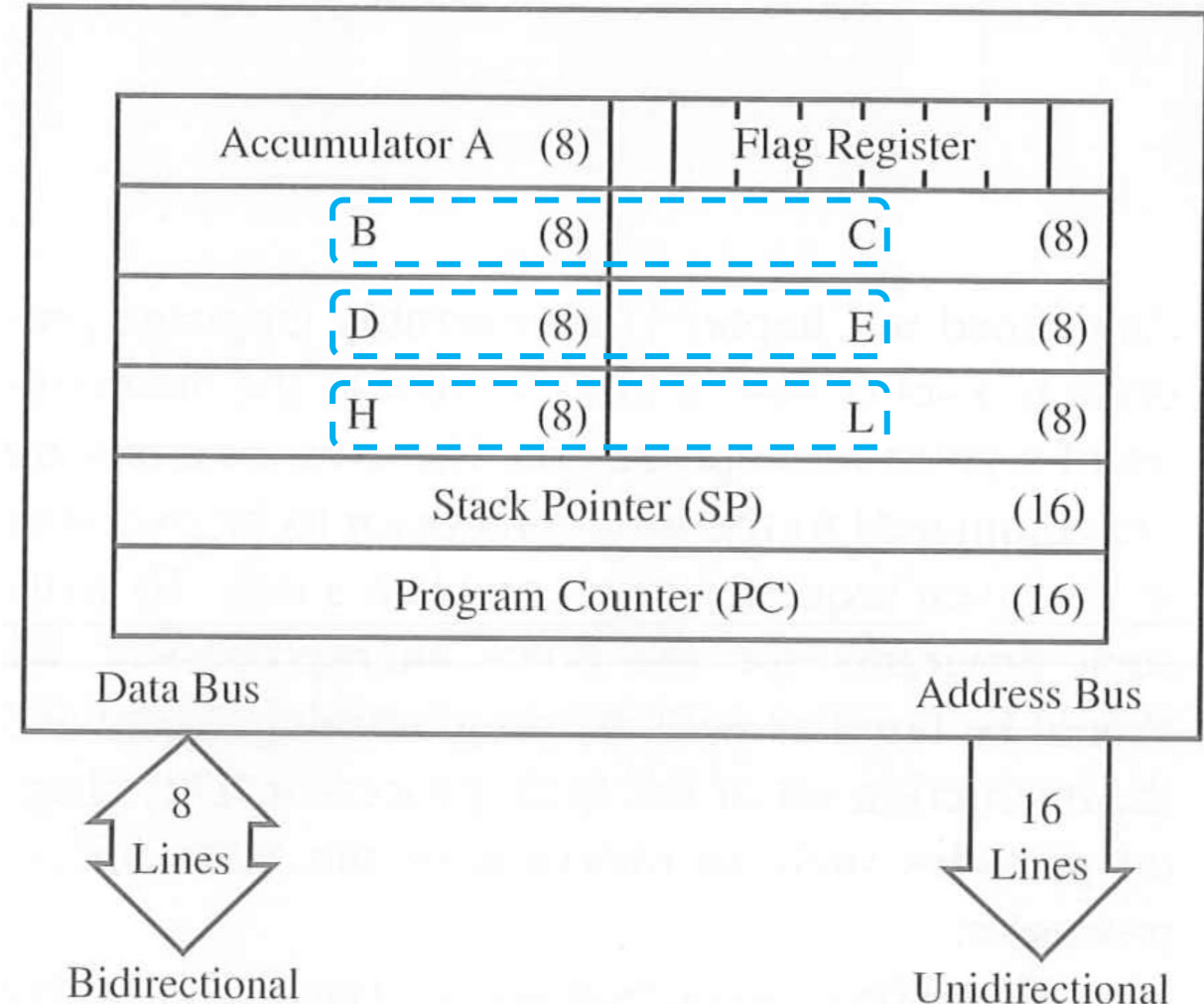
- Used for Direct Memory Access (DMA) signals.
- HOLD
 - ✓ External device requests bus access.
- HLDA
 - ✓ μ P acknowledges the HOLD request.



Intel 8085 Registers

An overview

- The 8085 is an **8-bit microprocessor**, so its ALU works mainly on **8-bit data**.
- Registers are very important temporary memory.
 - ✓ Fast access to frequently used data
 - ✓ Reduced memory traffic
 - ✓ Efficient instruction execution
 - ✓ Simplified programming with dedicated functions
- Total 10 registers **accessible** to programmers:
 - 6 **general purpose** registers (B, C, D, E, H, L)
 - 4 **special purpose** registers (A, F, PC, SP)
- Some **inaccessible** register are:
 - Memory Access Register (MAR)
 - Instruction Register (IR)
 - Temporary Register (W, Z)



Programmer's view of Intel 8085's Register Set

Intel 8085 Registers

Pairing up GPRs

- B-C, D-E, and H-L can be lumped together to work with 16 bits worth of data or addresses.

- For 16-bit **addresses**:

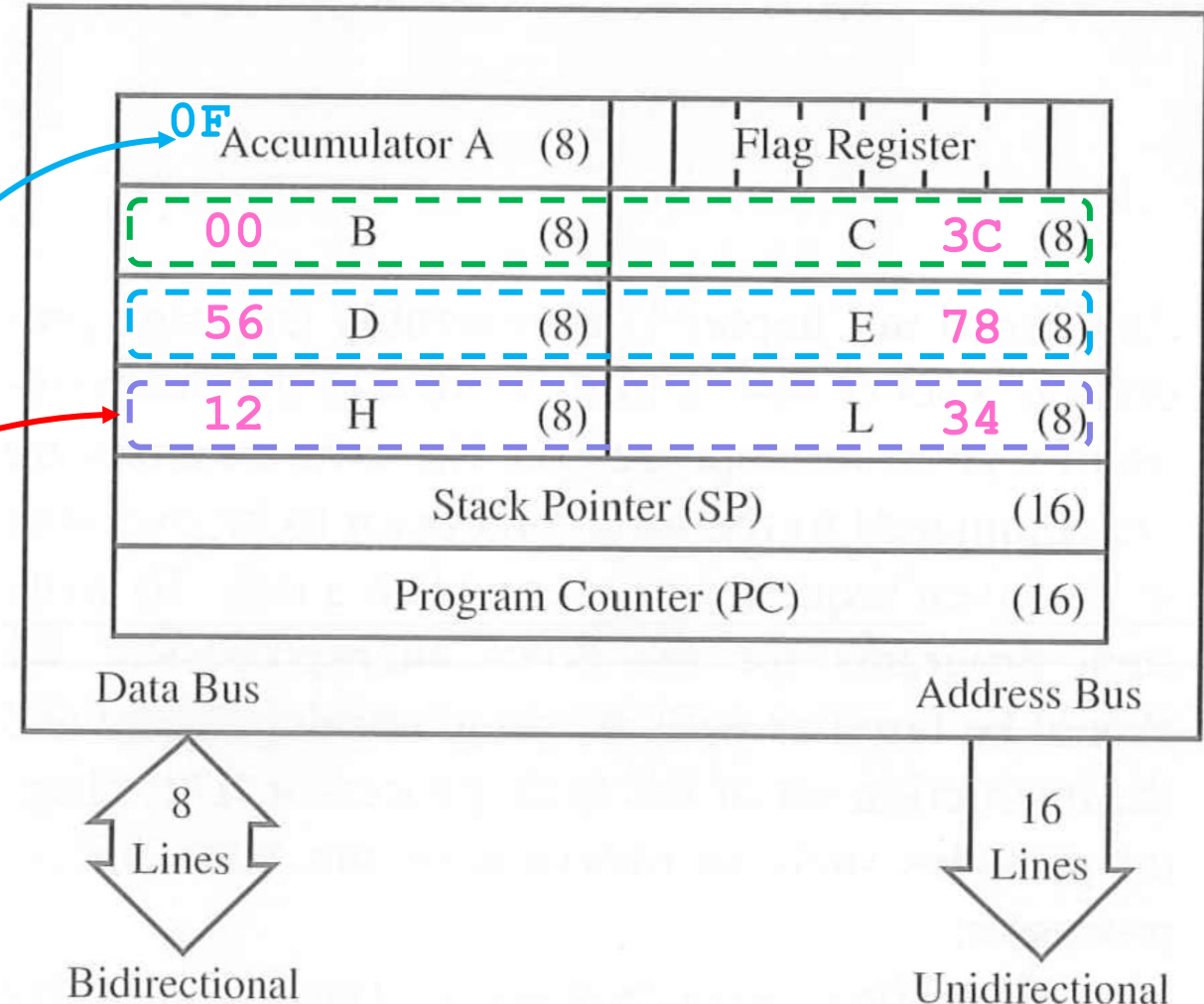
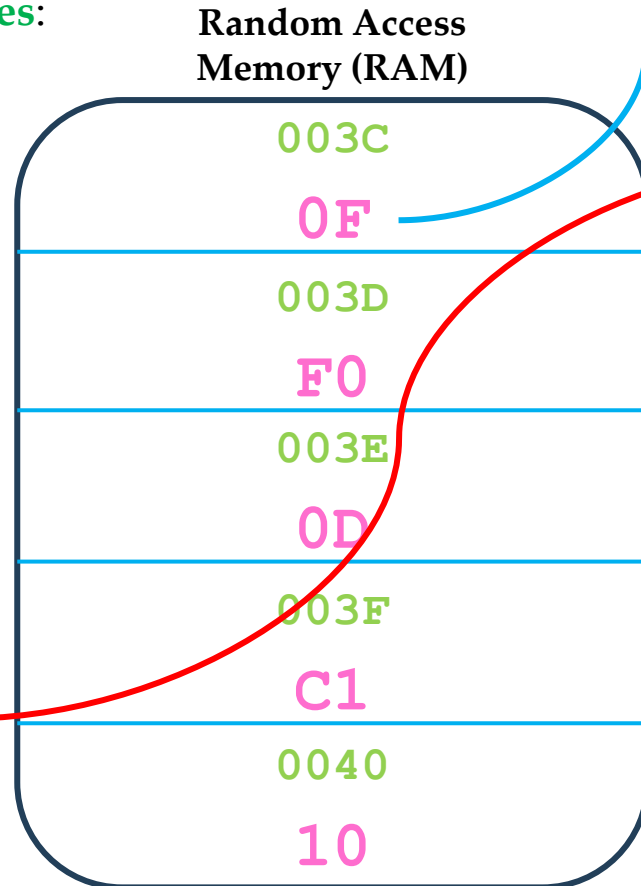
LDAX B

- For 16-bit **data**:

H-L acts as Acc.

DAD D

1	2	3	4	H
+	5	6	7	H
6	8	A	C	H



Programmer's view of Intel 8085's Register Set

Intel 8085 Registers

Implementing an “Index Register”

- Can be done using H-L.

MOV A, M → 003C

INX H

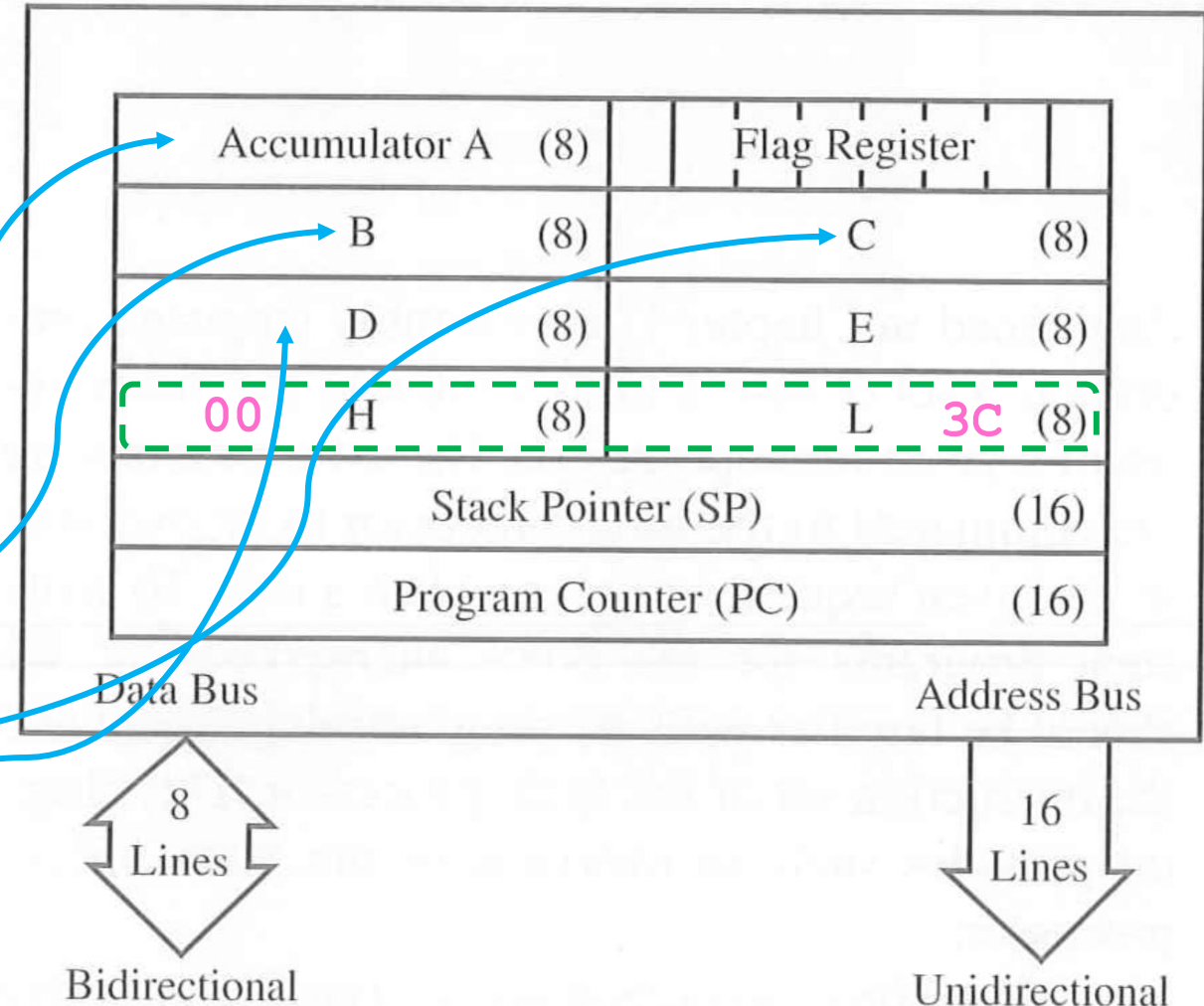
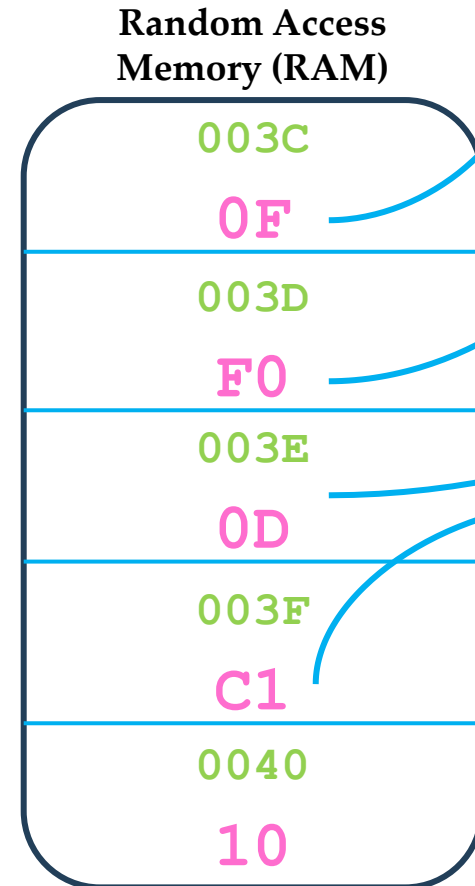
MOV B, M → 003D

INX H

MOV C, M → 003E

INX H

MOV D, M → 003F

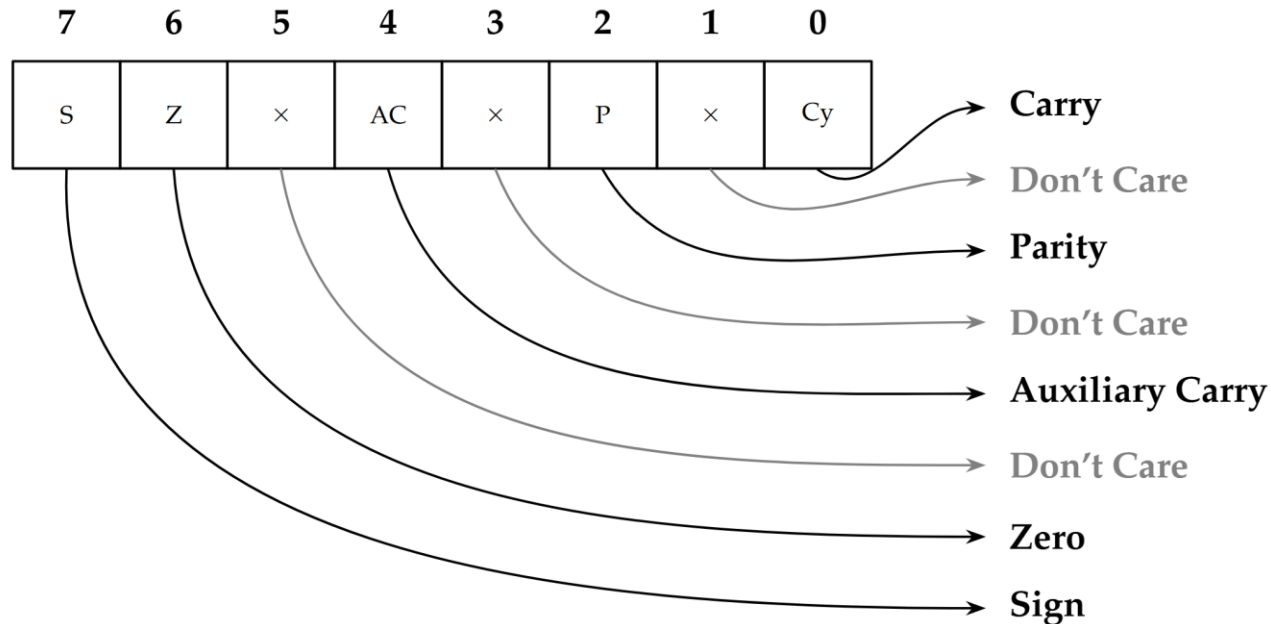


Programmer's view of Intel 8085's Register Set

Intel 8085 Registers

Status/Flag (F) Register

- 5 out of 8 bits serve their own unique purpose.



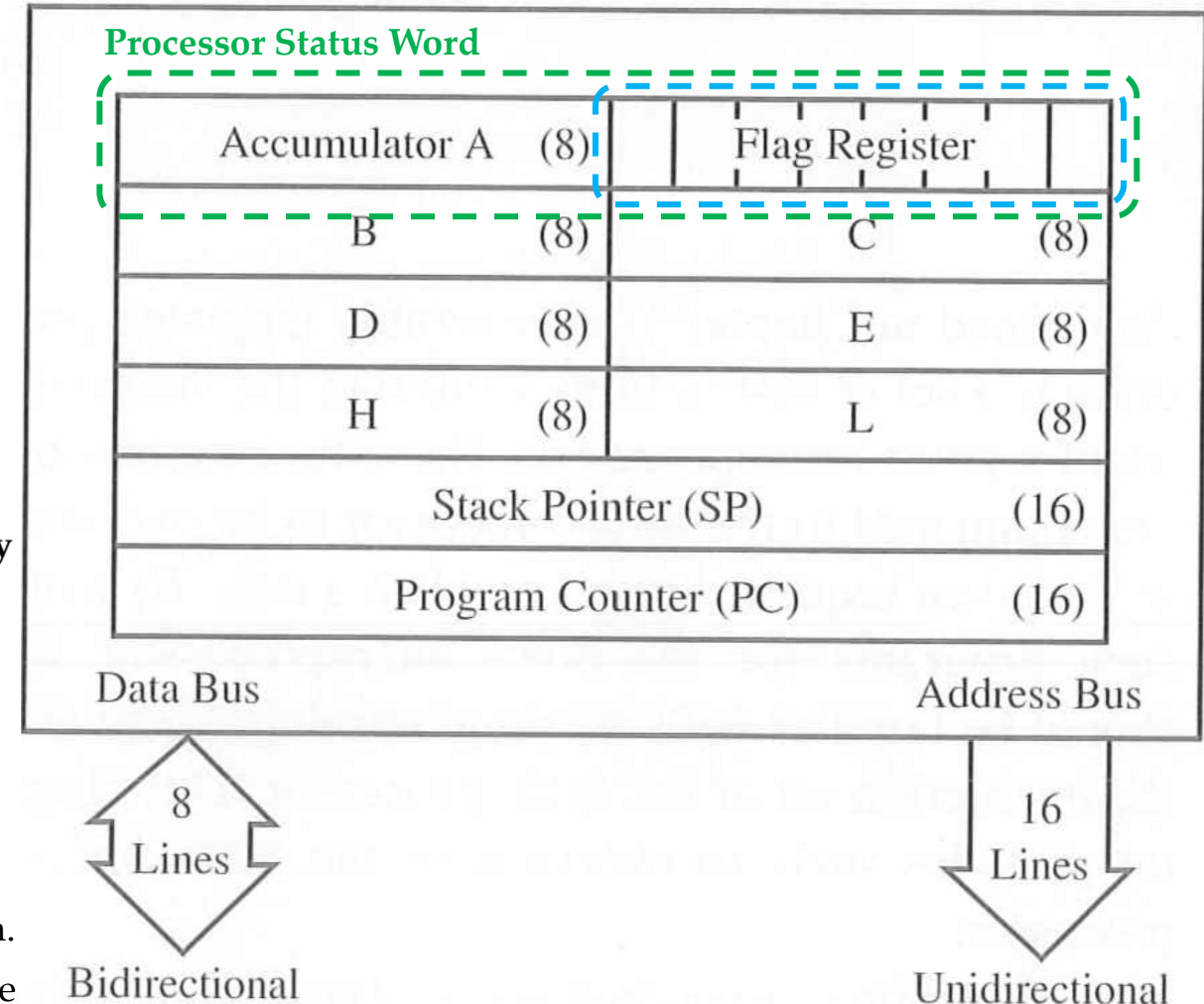
S: Set if the most significant bit of the result is 1.

Z: Set if the result of the operation is zero.

AC: Set if there is a carry from bit 3 to bit 4 during an arithmetic operation.

P: Set if the result contains an even number of 1s, that is, if the parity of the result is even.

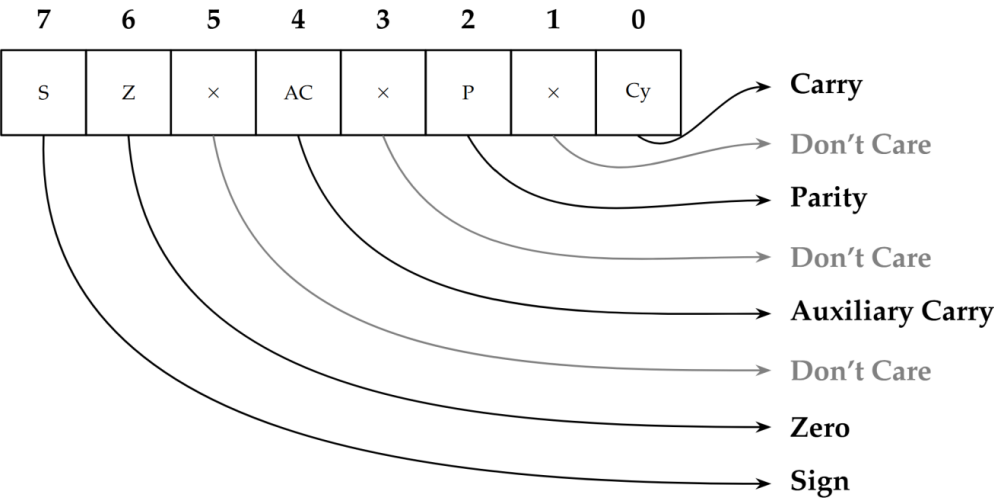
CY: Set if an addition produces a carry out of the most significant bit, or if a subtraction/comparison requires a borrow.



Programmer's view of Intel 8085's Register Set

Intel 8085 Registers

Status/Flag (F) Register with an example



```
MVI A, 32H
MVI B, 48H
ADD B
OUT 01H
HLT
```

$$32H + 48H = 7AH$$
$$\begin{array}{r} 0011\ 0010 \\ + 0100\ 1000 \\ \hline 0111\ 1010 \end{array}$$
$$A = 7AH$$

1. Sign Flag (S)

The sign flag depends on the most significant bit of the result.

$$7AH = 0111\ 1010$$

The most significant bit is 0, so the result is positive.

$S = 0$

2. Zero Flag (Z)

The zero flag is set only when the result is 00H.

Since

$$7AH \neq 00H$$

we get

$Z = 0$

3. Auxiliary Carry (AC)

Auxiliary carry checks whether there is a carry from bit 3 to bit 4.

Consider only the lower nibbles:

$$32H \rightarrow 0010$$

$$48H \rightarrow 1000$$

Now add:

$$0010 + 1000 = 1010$$

This produces no carry from bit 3 to bit 4.

$AC = 0$

4. Parity Flag (P)

The parity flag is set when the result contains an even number of 1s.

$$7AH = 0111\ 1010$$

Number of 1s in the result:

$$01111010$$

There are 5 ones, which is odd parity.

$P = 0$

5. Carry Flag (CY)

The carry flag is set if there is a carry out of the most significant bit.

From the binary addition:

$$\begin{array}{r} 0011\ 0010 \\ + 0100\ 1000 \\ \hline 0111\ 1010 \end{array}$$

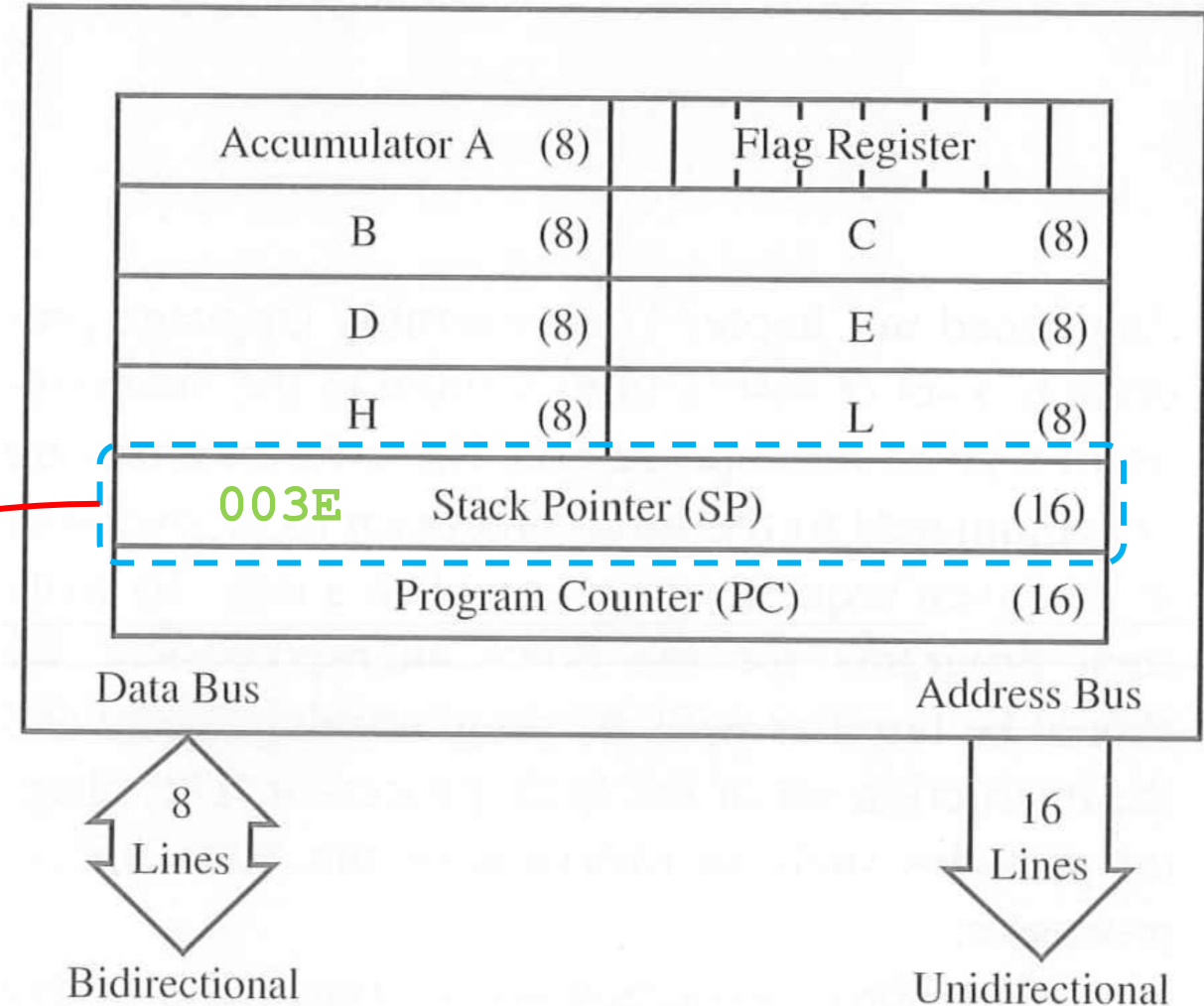
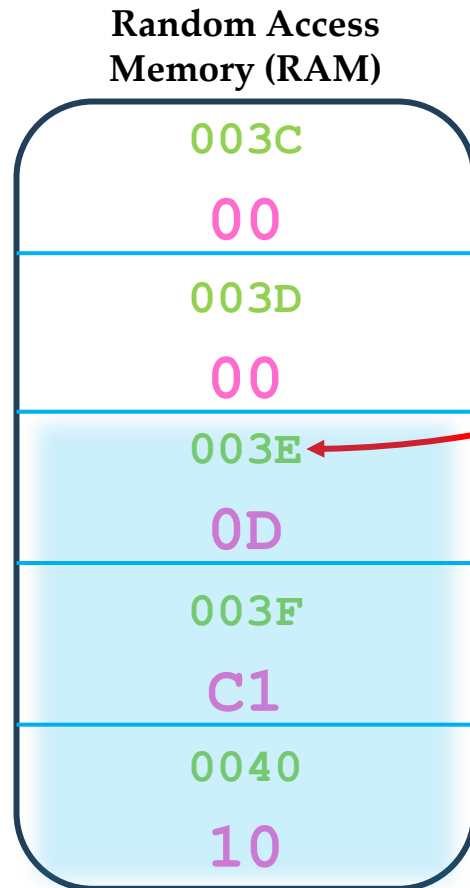
there is no carry beyond 8 bits.

$CY = 0$

Intel 8085 Registers

Stack Pointer (SP) Register

- Points to the top of the stack memory.



Programmer's view of Intel 8085's Register Set

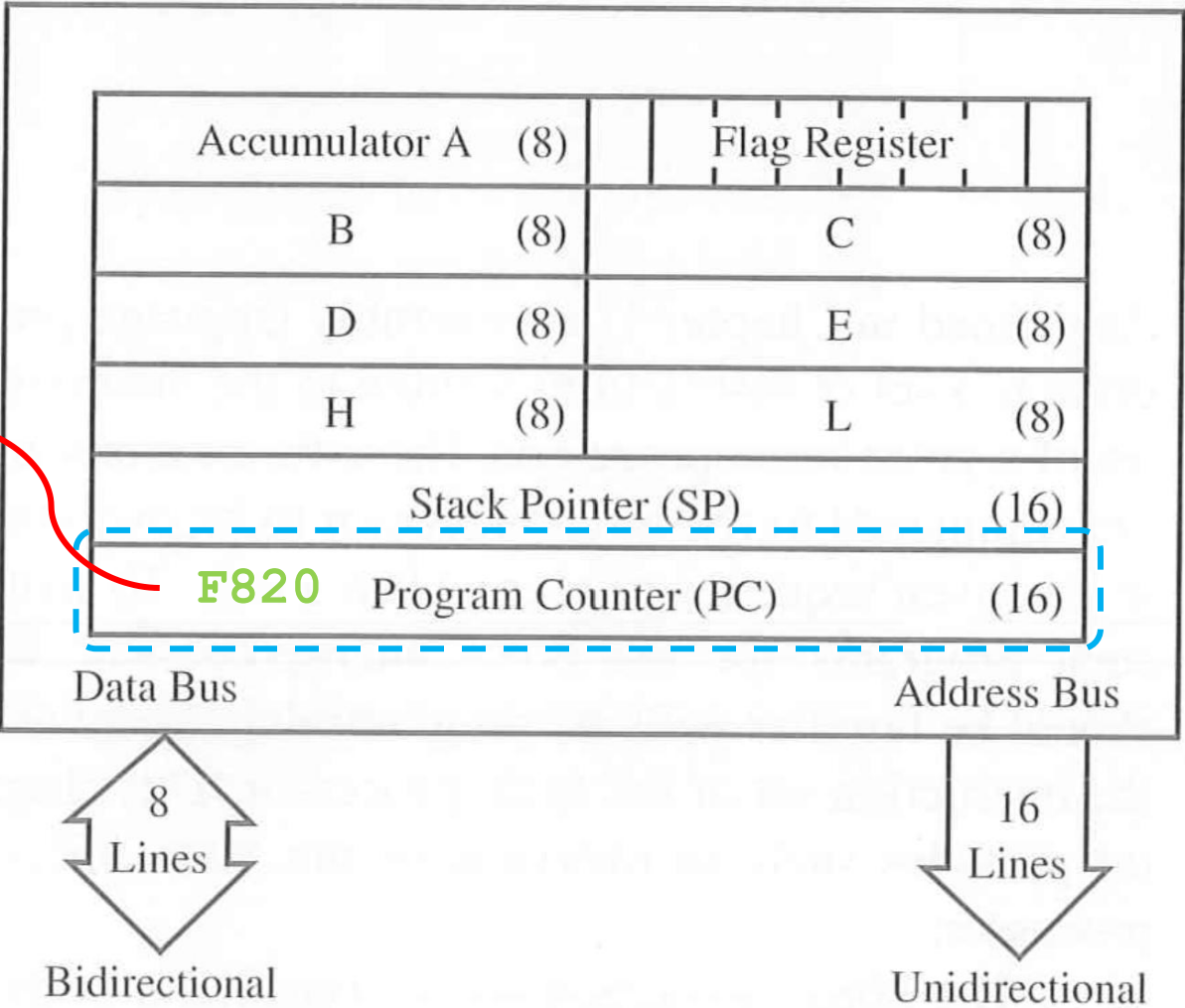
Intel 8085 Registers

Program Counter (PC) Register

- Points to address containing the next instruction to be executed and increments after each step.

```
LDA F840H
CMA
INR A
STA F850H
HLT
```

		0000
	:	:
LDA		F820
40		F821
F8		F822
CMA		F823
INR A		F824
STA		F825
50		F826
F8		F827
HLT		F828
:		:
		FFFF



Programmer's view of Intel 8085's Register Set